

Ground Control Witness Form & Diagrams

Project Name: Castello di Arco, Piazza III Novembre 3, 38062 Arco TN, Italy

Date: 30/10/2024

Project Code: ARCO

	Details	Notes/Comments	Justification/Explanation
Survey Overview			
Date:	30/10/2024	Date of survey	
Survey Conditions:	Indoors. Clear day, no previous rain.	Brief description of survey conditions.	Rain an important factor w.r.t. water ingress behind frescoes. Previous week of no rain!
No. of Scale Bars	4	Number of points being taken.	
Equipment:			
Model:	N/A	Make/model of surveying equipment used. This includes the make/model of any TST, prism, tripod	
Specifications:	N/A	Critical specifications only inc. accuracy, range etc.	
Additional Equipment	Tape Measure Thermal-specific survey markers	Equipment needed for the execution of the survey i.e. tape measure	Scale bars taken between survey markers
Survey Team:			
Planned / Set Up by:	Neil Sutherland Fabio Remondino	Lead individual responsible for survey planning	
Observed by:	Neil Sutherland Fabio Remondino	The individual taking the control point observations. This should remain consistent during the survey.	
Computed by:	Neil Sutherland	The team member responsible for the processing of the observations	

Notes:	
Notes:	Due to the nature of the frescoes, scale bars were configured in front of the façade with thermal-specific survey markers. Markers were attached to the barrier rope with blu-tac, with several markers rested against the wall to provide variable depth. Distances were measured between markers to provide scaling, Ground Control Scale Bars (GCSBs) and Check Scale Bars (CSBs) for the survey. Notably, there are no markers at the top of the wall (due to access and feasibility). Please refer to Img-Coords for specific details about thermal-specific survey marker locations.

Diagrams	
Site Diagram 1:	1. Representation of station/marker locations in relation to subject

Diagrams

Site Diagram 2:



P5 WAS NOT INCLUDED DUE TO MOVEMENT / MIS-ALIGNMENT



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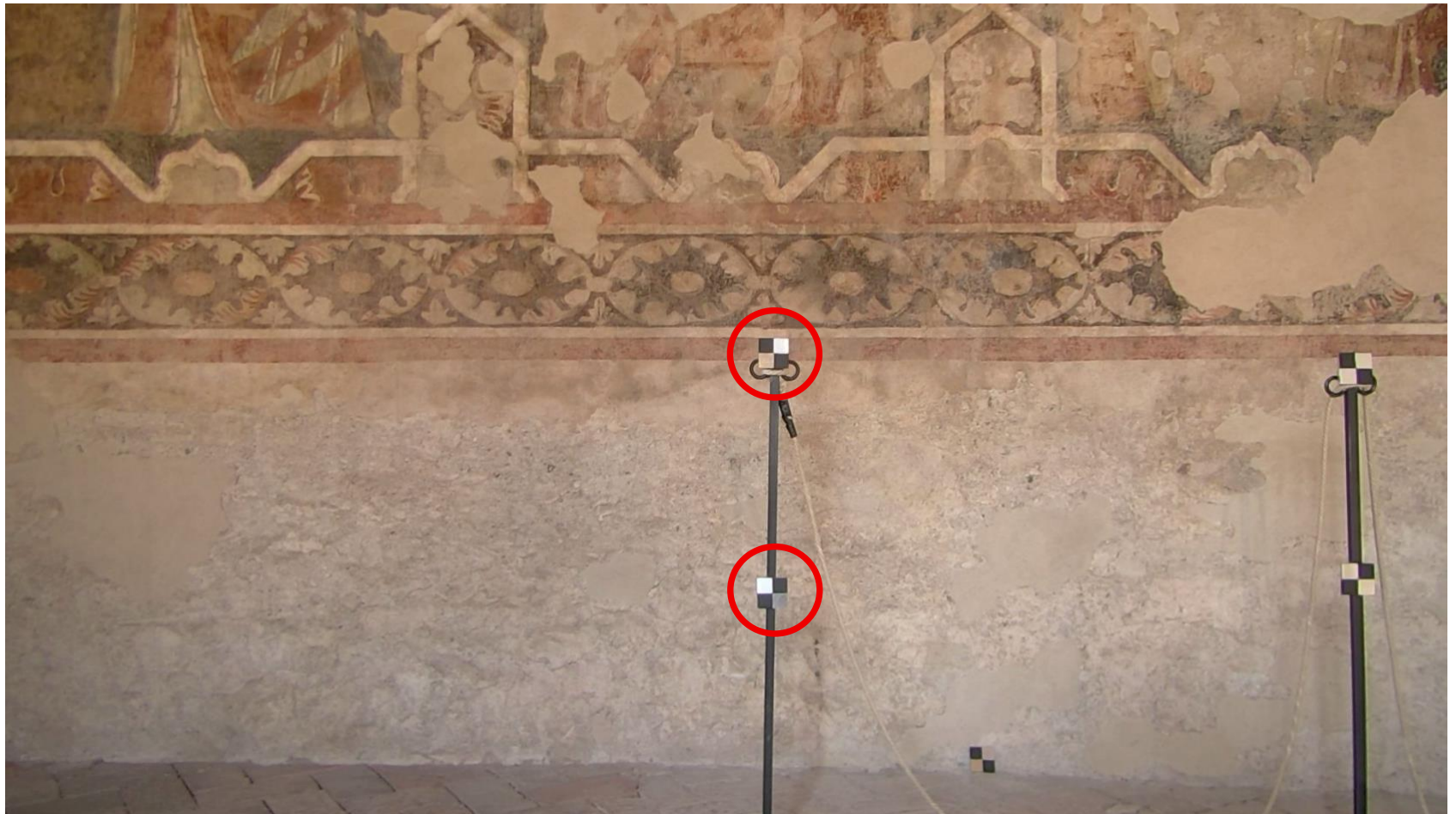
SCALE BAR 1

	Details	Notes/Comments	Justification/Explanation
Scale Bar Name:	P1_P2	Name saved within the equipment	
Distance (m):	0.480		
Accuracy (m):	0.005		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	

Diagrams

**Control Point
Photo(s):**



SCALE BAR 2

	Details	Notes/Comments	Justification/Explanation
Scale Bar Name:	P8_P10	Name saved within the equipment	
Distance (m):	0.830		
Accuracy (m):	0.005		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	

Diagrams

**Control Point
Photo(s):**



SCALE BAR 3

	Details	Notes/Comments	Justification/Explanation
Scale Bar Name:	P8_P11	Name saved within the equipment	
Distance (m):	0.957		
Accuracy (m):	0.005		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

Notes:	
Notes:	

Diagrams

**Control Point
Photo(s):**



SCALE BAR 4

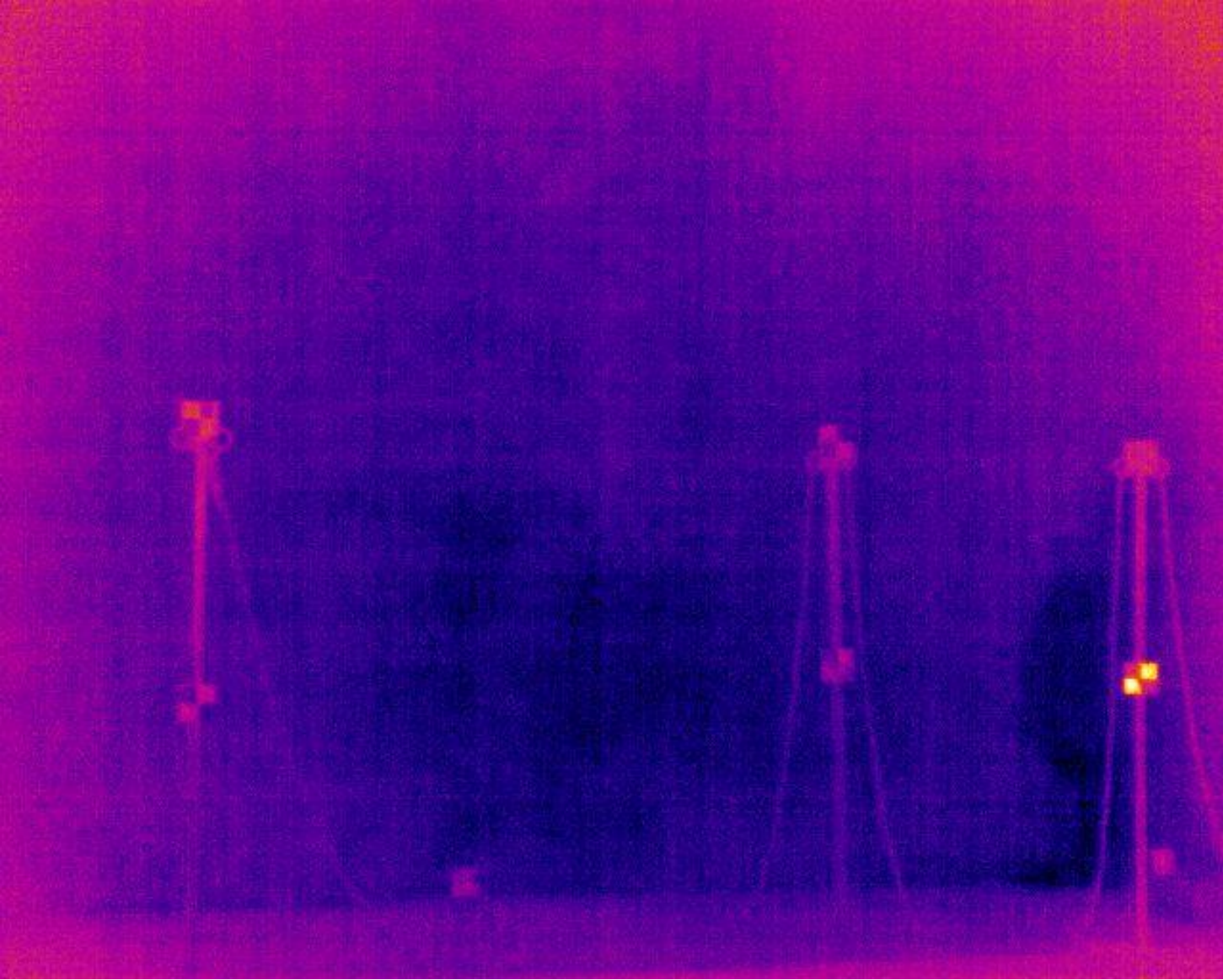
	Details	Notes/Comments	Justification/Explanation
Scale Bar Name:	P10_P11	Name saved within the equipment	
Distance (m):	0.476		
Accuracy (m):	0.005		
CRS / Datum:	N/A		
Station Name/No.:	N/A		

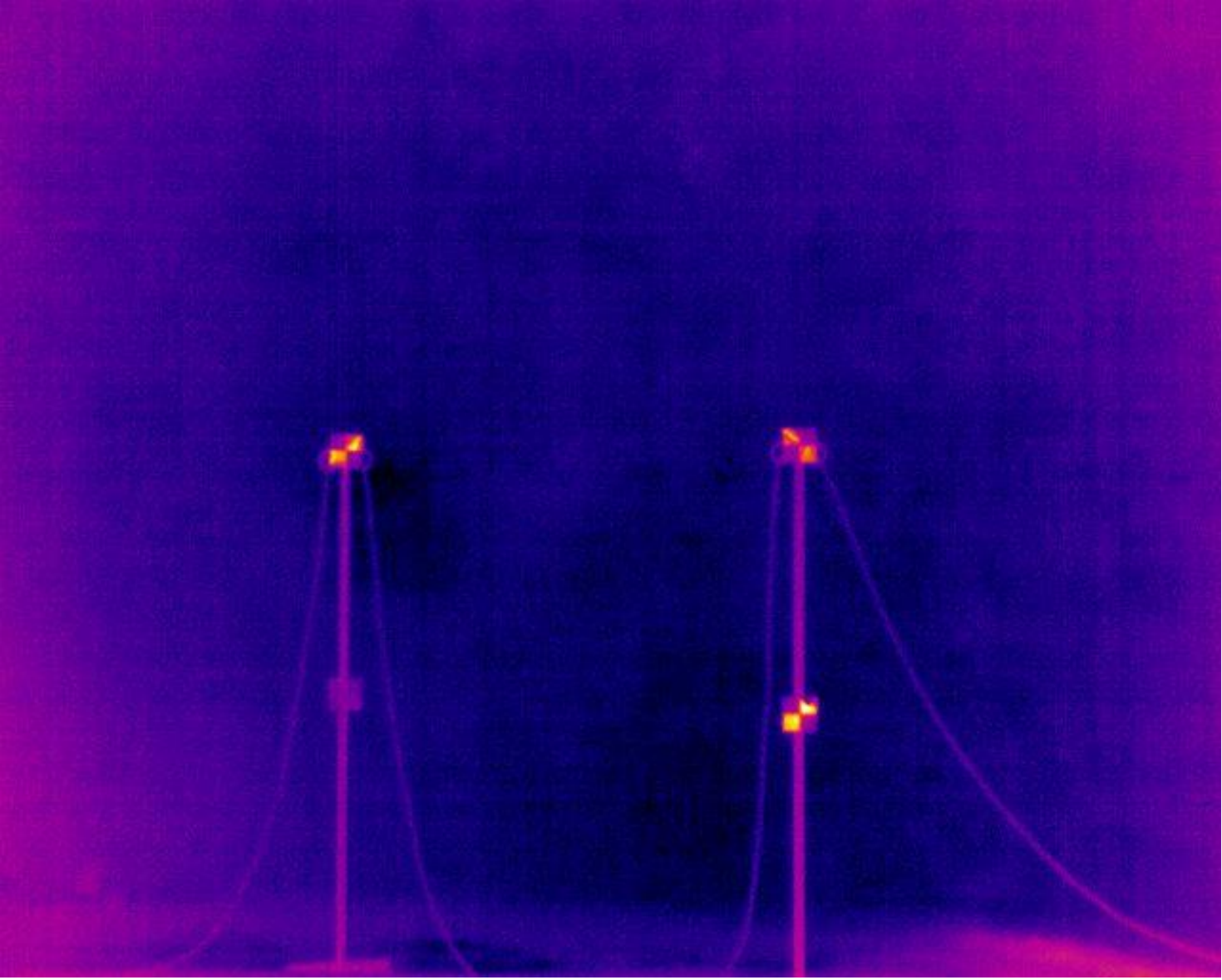
Notes:	
Notes:	

Diagrams

**Control Point
Photo(s):**



Diagrams	
Additional Images:	 A photograph of a construction site at night. Three tall, illuminated towers or cranes are visible, each with a bright light source at the top. The towers are positioned in a row, and the ground in the foreground is dark. The background is a dark, textured sky.

Diagrams	
Additional Images:	 A photograph showing two vertical poles, possibly part of a scientific experiment or a stage setup. Each pole has a bright red light at the top and a smaller red light or sensor at the bottom. The poles are connected by two red cables that curve outwards and then back in towards the bottom. The background is dark and textured, possibly a wall or a backdrop.